MIL-D-43597

18 September 1968

MILITARY SPECIFICATION

DRAWSHEET, MATTRESS, PROTECTIVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size cotton drawsheet.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
CCC-C-430	-	Cloth, Sheeting, Cotton, and Polyester and Cotton
CCC-T-191	-	Textile Test Methods
DDD-L-20	-	Label; For Clothing, Equipage, and Tentage, (General Use)
DDD-T-86	-	Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, Or Dyed)
PPP-B-636	-	Box, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

STANDARDS

FEDERAL

Fed. Std. No. 751 - Stitches, Seams and Stitchings

FSC 6532

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base
MS 16686	-	Medical Services Symbol

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-

3.1.1 Sheeting, cotton.- The cotton sheeting shall conform to type II, class 2, style B of CCC-C-430.

3.1.2 Tape, cotton.- The cotton tape used for binding shall conform to type I or II, class 2, 5/8 inch width of DDD-T-86.

3.1.3 Thread, cotton.- The cotton thread shall be white conforming to type IA2, ticket No. 70, 3 ply of V-T-276.

3.1.4 Rubber compound.- The elastomer used for combining the two plies of cotton sheeting shall be made from butyl rubber compounded to a shade which will not alter the shade of the fabric. The compound shall be nontoxic, non-irritant and free from objectionable odors.

3.2 Design.- The drawsheet shall consist of a large cotton sheet to which is laminated a smaller cotton sheet by means of a rubber elastomer. The edges of the smaller sheet shall be stitched to the large sheet across the length, and the widthwise edges of the entire drawsheet shall be covered with binding tape.

3.3 Construction.- The finished dimensions of the large sheet shall be a minimum of 70 inches wide and 45 $\pm$ 1/2 inches long. The smaller sheet shall be a minimum of 44-1/2 inches wide and of a length equal to the larger sheet. The laminated section shall thus be a minimum of 44-1/2 inches wide and shall be centered across the width dimension of the large sheet. The selvage edges of both sheets shall be parallel to each other across the length.

3.3.1 Lamination.- The laminated section of the drawsheet shall conform to the following requirements when tested as specified in 4.2.4.

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Weight of combined fabric, ounces per square yard.	12 to 15
Breaking strength, pounds, minimum	
Warp.	100
Fill.	90
Breaking strength, after accelerated aging, pounds, minimum	
Warp.	100
Fill.	90
Tearing strength, pounds, minimum	
Warp.	3.5
Fill.	3
Tearing strength, after accelerated aging, pounds, minimum	
Warp.	3.5
Fill.	3
Hydrostatic pressure, pounds per square inch (p.s.i.) minimum.	25
Hydrostatic pressure, after accelerated aging, (p.s.i.) minimum.	25
Hydrostatic pressure, after soaking in water, (p.s.i.) minimum.	25
Adhesion-fabric, pounds per inch, minimum.	7
Adhesion-fabric, after water soak, pounds per inch, minimum.	5
Adhesion-fabric, after accelerated aging, pounds per inch, minimum.	5
Stiffness, maximum.	12.5 centimeters.
Stiffness, after accelerated aging, maximum.	16 centimeters.

3.3.2 Stitches, seams and stitchings.- Stitches, seams and stitching types listed shall conform to Fed. Std. No. 751. All stitching shall be done with stitch type 301 and a minimum of 14 stitches per inch. The selvage edges of the smaller sheet shall be stitched to the larger sheet with seam type SSa-1, 1/16 to 1/8 inch from each edge. The binding tape shall cover both widthwise (cut or torn) edges and shall be stitched 1/16 to 1/8 inch from the edge of binding with seam type BSa-1.

3.3.2.1 Thread breaks and ends of seams.- The ends of all seams and stitchings when not caught in other seams and stitchings shall be backstitched or backtacked not less than 1/4 inch. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch.

3.4 Laundering.- The drawsheet shall be capable of withstanding five complete laundering cycles (washing and drying) when tested as specified in 4.2.4. The drawsheet shall show no signs of deterioration (delamination) at the completion of the laundering tests.

3.5 Label.- Each drawsheet shall be marked in any corner of the sheet. The marking shall conform to type IV, class 9 of DDD-L-20 and shall show fastness as specified for class 5 labels. The marking shall include the manufacturer's name in characters of approximately 1/4 inch. In addition, the marking shall include a caduceus approximately one inch in height centered between the letters "U" and "S" of appropriate size conforming to type I of MS 16686.

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3.6 Workmanship.- The finished drawsheets shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the supplier shall submit a certificate of compliance certifying that the requirements of 3.1.4 have been met.

4.2.2 Examination of the end item.- The classification of defects shall be in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one finished drawsheet. The lot size shall be expressed in units of one drawsheet. The inspection levels and acceptable quality levels shall be in accordance with 4.2.2.3.

4.2.2.1 Visual examination.- The finished drawsheet shall be visually examined for defects listed in table I.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Material	Hole, cut, tear or smash.	X	
	Floats or skips:		
	Multiple, 1/2 inch or more in length.	X	
	Multiple, less than 1/2 inch in length.		X
	Single, over 9 inches in length.		X
	Broken or missing yarns:		
	Three or more, less than 9 inches apart regardless of length.	X	
	Two, less than 9 inches apart over three inches in length.		X
	Hitchback:		
	Distorted, 2 square inches or more in area.	X	
	Distorted area less than 2 square inches and more than 1 square inch.		X
	Slub, slug, gout, jerked in filling or slough off:		
	Over 1/8 inch in thickness.	X	
	1/16 to 1/8 inch in thickness.		X
	Heavy place.		X
	Light place or thin place:		
	Three inches or less in longest dimension.		X
	More than 3 inches in longest dimension.	X	
	Temple cuts or abrasion marks.		X
Cleanness	Spot, stain or streak:		
	Exceeding 1 inch in longest dimension.	X	
	More than 1/4 inch but not exceeding 1 inch in longest dimension.		X
	Accumulation of spots or stains 1/4 inch or less in longest dimension (5 or more per sheet).		X
Seams and Stitching	Wrong seam type.	X	
	Wrong stitch type.	X	
	Needle chew.	X	
	Needle perforations.		X
	Stitch tension:		
	Loose, resulting in a loosely secured seam.		X
	Tight, stitches break when normal strain in applied to seam or stitching.		X

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and Stitching (cont'd)	Stitches per inch: One or two less than specified.		X
	Three or more less than specified.	X	
	Any open seam.	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Loose thread caught in stitching.		X
	Ends of seams and stitching (when not caught in other seams and stitching) backstitched or backtacked less than 1/4 inch.		X
	Thread breaks backstitched less than 1/2 inch.		X
Binding	Improperly or loosely applied.		X
	Does not cover both widthwise (cut or torn) edges.	X	
Components and assembly	Any component omitted, misplaced or not as specified or any operation omitted or not properly performed (unless otherwise classified herein).	X	
Marking	Omitted, incorrect, illegible, or misplaced.		X

4.2.2.2 Dimensional examination.- The drawsheet shall be examined for the dimensions specified in 3.3. Any measurement deviating from the specified dimension with tolerance shall be classified a defect.

4.2.2.3 Inspection levels and acceptable quality levels (AQLs).- The inspection levels and AQLs expressed in defects per hundred units shall be as follows:

Examination	Inspection level	AQL Major	Total
4.2.2.1	II	2.5	6.5
4.2.2.2	S-3	-	4.0

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4.2.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the following list. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, improper taping. Bulged or distorted container. Polyethylene bags heat sealed seams open, noncontinuous or crooked; vent hole missing or greater than 1/4 inch diameter. Number of intermediate packages per shipping container is more or less than required. Number of sheets per intermediate package is more or less than required. 1/

1/ For this defect, one intermediate package shall be examined from each shipping container in the sample.

4.2.4 Testing of the end item.- The test methods listed in table III shall be in accordance with CCC-T-191, except as otherwise noted, and as described herein. Unless otherwise specified, the physical and chemical values specified in section 3 represent the average of the number of test specimens selected for a sample unit for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of one drawsheet. The sample unit shall be two drawsheets, one for the laundering and marking tests and one for all other tests pertaining to the laminated section. The sample size shall be in accordance with table II and the lot shall be unacceptable if one or more units fail to meet any requirment specified.

TABLE II.- Sample size

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III.- End item tests

Characteristic	Requirement reference	Test method
Weight of combined fabric (oz./sq. yd.)	3.3.1	5041
Breaking strength:		
Warp	3.3.1	5100
Filling	3.3.1	5100
Breaking strength after accelerated aging:		
Warp	3.3.1	5100
Filling	3.3.1	5100
Tearing strength:		
Warp	3.3.1	5132
Filling	3.3.1	5132
Tearing strength after accelerated aging:		
Warp	3.3.1	5132
Filling	3.3.1	5132
Hydrostatic pressure	3.3.1	5512
Hydrostatic pressure after accelerated aging	3.3.1	5512
Hydrostatic pressure after soaking	3.3.1	5512 1/
Adhesion - fabric	3.3.1	5950
Adhesion - fabric after water soaking	3.3.1	5950 1/
Adhesion - fabric after accelerated aging	3.3.1	5950
Stiffness	3.3.1	5204
Stiffness after accelerated aging	3.3.1	5204

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TABLE III.- End item tests (cont'd)

Characteristic	Requirement reference	Test method
Accelerated aging	3.3.1	5852 2/
Laundering	3.4	4.3.1

1/ The sample shall be immersed in water at a temperature of 24° ± 2.8° centigrade (75° ± 5° fahrenheit) for two hours, removed from the water, and while dripping wet, tested by the applicable procedures.

2/ The exposure period shall be 96 hours.

4.3 Test.-

4.3.1 Laundering test.-

4.3.1.1 Apparatus.- The apparatus shall consist of a wash wheel and extractor as specified in method 5550 of CCC-T-191.

4.3.1.2 Procedure.-

4.3.1.2.1 Laundering.- The laundering procedure shall be in accordance with 4.1, method 5518 of CCC-T-191, except that the first water operation (Operation No. 3) shall be replaced by a bleach operation, as follows:

Operation	Composition of washing bath	Level in wheel (Inches)	Temperature	Running time (Minutes)
3	Bleach, 2 oz., 1% solution	6	150°F	8

Five water operations, and the remainder of the cycle as specified in method 5518, shall follow the bleach operation. After the laundering cycle, the specimen shall be extracted for 5 to 7 minutes using the extractor specified in 4.3.1.1.

4.3.1.2.2 Drying.- The drawsheet shall be dried by one of the following methods:

a. Tumble drying. The drawsheet shall be placed in a tumble dryer, until dry. The temperature shall be maintained at 105° to 110°C. (221° to 230°F.).

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b. Drying oven. The drawsheet shall be spread out on drying trays. The trays shall be placed in a circulating air oven, thermostatically controlled and capable of maintaining the temperature at 105° to 110°C. (221° to 230°F.), until dry.

4.3.1.2.3 Pressing.- The drawsheet shall be pressed as specified in method 5550 of CCC-T-191.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit packaging.- Each drawsheet shall be neatly folded to approximate 12 by 9 inches. Each sheet shall then be inserted in a close-fitting flat style clear polyethylene film bag of 0.00125-inch thickness (± 20 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final heat seal closure to the bag shall be made as close as possible to the open end. A 1/4-inch maximum diameter hole shall be made at one corner of each polyethylene bag to allow the excess air to escape.

5.1.1.2 Intermediate packaging.- Six drawsheets, unit packaged as specified in 5.1.1.1, shall be positioned flat in a fiberboard box conforming to style RSC, type CF (variety SW), class domestic, grade 125 of PPP-B-636. Inside dimensions of each box shall approximate 12 inches in length, 9 inches in width and 5-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Level C.- Each drawsheet shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Forty-eight drawsheets, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A packages shall be packed flat two in length, two in width and two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 25 inches in length, 19 inches in width and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

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5.2.2 Level B.- Forty-eight drawsheets, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety DW) or SF, class domestic, grade 350 of PPP-B-636. Level A packages shall be packed flat two in length, two in width and two in depth within a shipping container. Inside dimension of each shipping container shall approximate 25 inches in length, 19 inches in width and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Drawsheets, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Palletization.- Unless otherwise specified (see 6.2), drawsheets packed as specified in 5.2, shall be palletized in accordance with Load Type I of MIL-STD-147, except that for overseas shipment the over-all height of the load shall not exceed 43-inches. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Container patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the patterns of each coarse. If the container is of a size which does not conform to any of the container patterns specified in MIL-STD-147, the container pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4.1 Polyethylene bagged packages.- Polyethylene bagged packages shall have the required information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
QUANTITY

6. NOTES

6.1 Intended use.- The drawsheets covered by this specification are intended to be used to protect hospital mattresses from body fluids.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(d) When palletization is not required (see 5.3).

Custodians:

Army - GL
Navy - MS
Air Force - 03

Preparing activity:

Army - GL

Project No. 6532-1012

Review activities:

Army - MD
Navy - MS
Air Force - 03
DSA - DM

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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